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C9: Confidence interval for the mean when variance is unknown

P1 Problem description A ride-sharing platform studies daily gross driver payouts (hundreds of USD) in one urban zone. A random sample of $n = 10$ days gives the raw data:

$$42,\ 38,\ 45,\ 40,\ 41,\ 39,\ 44,\ 43,\ 37,\ 41$$

1) Construct a 95% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean daily loan disbursement}, \quad n = 10, \quad \bar{x} = \frac{410}{10} = 41.00$$

$$\sum (x_i - \bar{x})^2 = 60, \quad s = \sqrt{\frac{60}{9}} = 2.582, \quad df = n - 1 = 9$$

$$t^* = t_{0.025,9} \approx 2.262, \quad SE = \frac{s}{\sqrt{n}} = \frac{2.582}{\sqrt{10}} = 0.816$$

$$E_t = t^*(SE) = 2.262(0.816) = 1.85$$
$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 41.00 \pm 2.262 \left(\frac{2.582}{\sqrt{10}}\right) = 41.00 \pm 1.85$$
$$\mu \in [39.15,\ 42.85]$$

Interpretation: With 95% confidence, the zone's mean daily driver payout is between 39.15 and 42.85 hundred USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.025} = 1.960$$

$$E_z = z^*(SE) = 1.960(0.816) = 1.60$$

$$\mu \in 41.00 \pm E_z = 41.00 \pm 1.60 = [39.40,\ 42.60]$$

C9: Structured Decision Procedure

$$t^* = 2.262 > 1.960 = z^*$$
$$E_t = 1.85, \quad E_z = 1.60$$
$$\text{Width}_t = 2(1.85) = 3.70, \qquad \text{Width}_z = 2(1.60) = 3.20$$
$$\text{Absolute difference} = 3.70 - 3.20 = 0.50$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{3.70 - 3.20}{3.20} \times 100 = 15.63\%$$

Because $df = 9$, the t critical value is noticeably larger; therefore the t-interval is wider. Final decision: the t-interval is the appropriate exact interval under unknown variance and is 15.63% wider than the z-interval.

P2 Problem description An investment app analyzes mean per-trade platform fees (USD) paid by users in their 20s. A random sample of $n = 18$ transactions gives the raw data:

$$1.8,\ 2.1,\ 1.9,\ 2.4,\ 2.0,\ 1.7,\ 2.2,\ 2.3,\ 1.8,\ 2.5,\ 2.1,\ 1.9,\ 2.0,\ 2.2,\ 1.6,\ 2.4,\ 2.3,\ 1.9$$

1) Construct a 90% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean transaction fee}, \quad n = 18, \quad \bar{x} = \frac{37.1}{18} = 2.061$$

$$\sum (x_i - \bar{x})^2 = 1.143, \quad s = \sqrt{\frac{1.143}{17}} = 0.259, \quad df = 17$$

$$t^* = t_{0.05,17} \approx 1.740, \quad SE = \frac{0.259}{\sqrt{18}} = 0.0611$$

$$E_t = t^*(SE) = 1.740(0.0611) = 0.106$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 2.061 \pm 1.740 \left(\frac{0.259}{\sqrt{18}}\right) = 2.061 \pm 0.106$$

$$\mu \in [1.955,\ 2.167]$$

Interpretation: With 90% confidence, the app's mean per-trade fee is between 1.955 and 2.167 USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.05} = 1.645$$

$$E_z = z^*(SE) = 1.645(0.0611) = 0.101$$

$$\mu \in 2.061 \pm E_z = 2.061 \pm 0.101 = [1.960,\ 2.162]$$

C9: Structured Decision Procedure

$$t^* = 1.740 > 1.645 = z^*$$

$$E_t = 0.106, \quad E_z = 0.101$$

$$\text{Width}_t = 0.212, \qquad \text{Width}_z = 0.202$$

$$\text{Absolute difference} = 0.212 - 0.202 = 0.010$$

$$\text{Relative difference (base } z\text{-interval)} = \frac{0.212 - 0.202}{0.202} \times 100 = 4.95\%$$

At $df = 17$, t remains heavier-tailed than z, so the t-interval is wider. Final decision: use the t-interval as exact under unknown variance; it is 4.95% wider than the z-interval.

P3 Problem description A food-delivery app studies mean order value (USD) during weekday mornings for young professionals. A random sample of $n = 30$ receipts gives the raw data:

$$115,\ 122,\ 118,\ 130,\ 126,\ 119,\ 121,\ 124,\ 128,\ 117, 123,\ 120,\ 125,\ 129,\ 116,\ 127,\ 122,$$

$$118,\ 124,\ 121, 126,\ 119,\ 123,\ 128,\ 120,\ 125,\ 117,\ 124,\ 122,\ 126$$

1) Construct a 99% confidence interval for the population mean using the t-distribution. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean basket value}, \quad n = 30, \quad \bar{x} = \frac{3675}{30} = 122.50$$

$$\sum(x_i - \bar{x})^2 = 477.5, \quad s = \sqrt{\frac{477.5}{29}} = 4.058, \quad df = 29$$

$$t^* = t_{0.005,29} \approx 2.756, \quad SE = \frac{4.058}{\sqrt{30}} = 0.741$$

$$E_t = t^*(SE) = 2.756(0.741) = 2.04$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 122.50 \pm 2.756 \left(\frac{4.058}{\sqrt{30}}\right) = 122.50 \pm 2.04$$

$$\mu \in [120.46,\ 124.54]$$

Interpretation: With 99% confidence, the platform's mean weekday-morning order value is between 120.46 and 124.54 USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.005} = 2.576$$

$$E_z = z^*(SE) = 2.576(0.741) = 1.91$$

$$\mu \in 122.50 \pm E_z = 122.50 \pm 1.91 = [120.59,\ 124.41]$$

C9: Structured Decision Procedure

$$t^* = 2.756 > 2.576 = z^*$$

$$E_t = 2.04, \quad E_z = 1.91$$

$$\text{Width}_t = 4.08, \qquad \text{Width}_z = 3.82$$

$$\text{Absolute difference} = 4.08 - 3.82 = 0.26$$

$$\text{Relative difference (base } z\text{-interval)} = \frac{4.08 - 3.82}{3.82} \times 100 = 6.81\%$$

At moderate n, t and z are closer, but t is still wider and remains the exact method. Final decision: the t-interval is selected and is 6.81% wider than the z-interval.

P4 Problem description A gig-delivery startup studies average delivery time (minutes) for same-city orders. For $n = 60$ orders, data are grouped as follows:

Class (minutes)	Midpoint m_i	f_i
20–24	22	6
25–29	27	9
30–34	32	14
35–39	37	13
40–44	42	10
45–49	47	8

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean delivery time (minutes)}, \quad n = \sum f_i = 60, \quad \sum f_i m_i = 2100, \quad \sum f_i m_i^2 = 77910$$

$$\bar{x} \approx \frac{\sum f_i m_i}{n} = \frac{2100}{60} = 35.00$$

$$s \approx \sqrt{\frac{\sum f_i m_i^2 - n\bar{x}^2}{n-1}} = \sqrt{\frac{77910 - 60(35)^2}{59}} = 7.602, \quad df = 59$$

$$t^* = t_{0.025,59} \approx 2.000, \quad SE = \frac{7.602}{\sqrt{60}} = 0.981$$

$$E_t = t^*(SE) = 2.000(0.981) = 1.96$$

$$\bar{x} \pm t^*\left(\frac{s}{\sqrt{n}}\right) = 35.00 \pm 2.000\left(\frac{7.602}{\sqrt{60}}\right) = 35.00 \pm 1.96$$

$$\mu \in [33.04,\ 36.96]$$

Interpretation: With 95% confidence, mean same-city delivery time is between 33.04 and 36.96 minutes.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.025} = 1.960$$

$$E_z = z^*(SE) = 1.960(0.981) = 1.92$$

$$\mu \in 35.00 \pm E_z = 35.00 \pm 1.92 = [33.08,\ 36.92]$$

C9: Structured Decision Procedure

$$t^* = 2.000 > 1.960 = z^*$$

$$E_t = 1.96, \quad E_z = 1.92$$

$$\text{Width}_t = 3.92, \qquad \text{Width}_z = 3.84$$

$$\text{Absolute difference} = 3.92 - 3.84 = 0.08$$

$$\text{Relative difference (base } z\text{-interval)} = \frac{3.92 - 3.84}{3.84} \times 100 = 2.08\%$$

With $df = 59$, convergence is visible, but the t-interval remains slightly wider. Final decision: retain the t-interval as exact under unknown variance; it is 2.08% wider than the z-interval.

P5 Problem description A direct-to-consumer e-commerce startup estimates mean daily ad spending (thousand USD). For $n = 120$ days, grouped data are:

Class (thousand USD)	m_i	f_i
50–55	52.5	12
55–60	57.5	18
60–65	62.5	28
65–70	67.5	24
70–75	72.5	22
75–80	77.5	16

1) Construct a 90% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean daily ad spending}, \quad n = 120, \quad \sum f_i m_i = 7870, \quad \sum f_i m_i^2 = 524975$$

$$\bar{x} \approx \frac{7870}{120} = 65.583$$

$$s \approx \sqrt{\frac{524975 - 120(65.583)^2}{119}} = 7.620, \quad df = 119$$

$$t^* = t_{0.05,119} \approx 1.658, \quad SE = \frac{7.620}{\sqrt{120}} = 0.696$$

$$E_t = t^*(SE) = 1.658(0.696) = 1.15$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 65.583 \pm 1.658 \left(\frac{7.620}{\sqrt{120}}\right) = 65.583 \pm 1.15$$

$$\mu \in [64.43,\ 66.74]$$

Interpretation: With 90% confidence, mean daily ad spending is between 64.43 and 66.74 thousand USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.05} = 1.645$$

$$E_z = z^*(SE) = 1.645(0.696) = 1.14$$

$$\mu \in 65.583 \pm E_z = 65.583 \pm 1.14 = [64.44,\ 66.72]$$

C9: Structured Decision Procedure

$$t^* = 1.658 > 1.645 = z^*$$

$$E_t = 1.15, \quad E_z = 1.14$$

$$\text{Width}_t = 2.30, \qquad \text{Width}_z = 2.28$$

$$\text{Absolute difference} = 2.30 - 2.28 = 0.02$$

$$\text{Relative difference (base } z\text{-interval)} = \frac{2.30 - 2.28}{2.28} \times 100 = 0.88\%$$

At $df = 119$, t and z are nearly equal, but t remains slightly wider. Final decision: use the t-interval as exact under unknown variance; it is 0.88% wider than the z-interval.

P6 Problem description A subscription software network studies mean monthly recurring revenue (thousand USD) for small creator-led apps. For $n = 250$ firms, grouped data are:

Class (thousand USD)	m_i	f_i
80–90	85	18
90–100	95	32
100–110	105	46
110–120	115	58
120–130	125	44
130–140	135	30
140–150	145	22

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean monthly recurring revenue}, \quad n = 250, \quad \sum f_i m_i = 28810, \quad \sum f_i m_i^2 = 3403450$$

$$\bar{x} \approx \frac{28810}{250} = 115.24$$

$$s \approx \sqrt{\frac{3403450 - 250(115.24)^2}{249}} = 16.741, \quad df = 249$$

$$t^* = t_{0.025,249} \approx 1.969, \quad SE = \frac{16.741}{\sqrt{250}} = 1.059$$

$$E_t = t^*(SE) = 1.969(1.059) = 2.08$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}} \right) = 115.24 \pm 1.969 \left(\frac{16.741}{\sqrt{250}} \right) = 115.24 \pm 2.08$$

$$\mu \in [113.16,\ 117.32]$$

Interpretation: With 95% confidence, mean monthly recurring revenue is between 113.16 and 117.32 thousand USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.025} = 1.960$$

$$E_z = z^*(SE) = 1.960(1.059) = 2.08$$

$$\mu \in 115.24 \pm E_z = 115.24 \pm 2.08 = [113.16,\ 117.32]$$

C9: Structured Decision Procedure

$$t^* = 1.969 > 1.960 = z^*$$

$$E_t = 2.08, \quad E_z = 2.08$$

$$\text{Width}_t = 4.16, \qquad \text{Width}_z = 4.16$$

$$\text{Absolute difference} = 4.16 - 4.16 = 0.00 \text{ (more precisely } 0.02\text{)}$$

$$\text{Relative difference (base } z\text{-interval)} \approx \frac{0.02}{4.16} \times 100 = 0.48\%$$

For large df, practical differences are tiny, but the t-interval is still (slightly) wider. Final decision: keep the t-interval as exact under unknown variance; it is about 0.48% wider than the z-interval.

P7 Problem description A fintech savings app estimates the mean monthly account balance (hundreds of USD) of active users in their 20s. For $n = 500$ users, grouped data are:

Class (hundreds of USD)	m_i	f_i
60–65	62.5	30
65–70	67.5	52
70–75	72.5	84
75–80	77.5	108
80–85	82.5	96
85–90	87.5	68
90–95	92.5	40
95–100	97.5	22

1) Construct a 99% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean monthly balance}, \quad n = 500, \quad \sum f_i m_i = 39560, \quad \sum f_i m_i^2 = 3174700$$

$$\bar{x} \approx \frac{39560}{500} = 79.12$$

$$s \approx \sqrt{\frac{3174700 - 500(79.12)^2}{499}} = 8.924, \quad df = 499$$

$$t^* = t_{0.005,499} \approx 2.586, \quad SE = \frac{8.924}{\sqrt{500}} = 0.399$$

$$E_t = t^*(SE) = 2.586(0.399) = 1.03$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 79.12 \pm 2.586 \left(\frac{8.924}{\sqrt{500}}\right) = 79.12 \pm 1.03$$

$$\mu \in [78.09,\ 80.15]$$

Interpretation: With 99% confidence, the app's mean monthly balance is between 78.09 and 80.15 hundred USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.005} = 2.576$$

$$E_z = z^*(SE) = 2.576(0.399) = 1.03$$

$$\mu \in 79.12 \pm E_z = 79.12 \pm 1.03 = [78.09,\ 80.15]$$

C9: Structured Decision Procedure

$$t^* = 2.586 > 2.576 = z^*$$

$$E_t = 1.03, \quad E_z = 1.03$$

$$\text{Width}_t = 2.06, \qquad \text{Width}_z = 2.06$$

$$\text{Absolute difference} = 2.06 - 2.06 = 0.00 \text{ (more precisely } 0.01)$$

$$\text{Relative difference (base } z\text{-interval)} \approx \frac{0.01}{2.06} \times 100 = 0.49\%$$

With hundreds of observations, t and z are nearly indistinguishable numerically. Final decision: the t-interval remains the exact choice and is about 0.49% wider than the z-interval.

P8 Problem description A national online tutoring marketplace estimates mean annual instructor earnings (USD) for active tutors. For $n = 1000$ policies, grouped data are:

Class (USD)	m_i	f_i
125–135	130	70
135–145	140	130
145–155	150	220
155–165	160	250
165–175	170	180
175–185	180	100
185–195	190	50

1) Construct a 95% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$\mu =$ population mean annual instructor earnings, $\quad n = 1000, \quad \sum f_i m_i = 158400, \quad \sum f_i m_i^2 = 25396000$

$$\bar{x} \approx \frac{158400}{1000} = 158.40$$

$$s \approx \sqrt{\frac{25396000 - 1000(158.4)^2}{999}} = 15.417, \quad df = 999$$

$$t^* = t_{0.025,999} \approx 1.962, \quad SE = \frac{15.417}{\sqrt{1000}} = 0.488$$

$$E_t = t^*(SE) = 1.962(0.488) = 0.96$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 158.40 \pm 1.962 \left(\frac{15.417}{\sqrt{1000}}\right) = 158.40 \pm 0.96$$

$$\mu \in [157.44,\ 159.36]$$

Interpretation: With 95% confidence, mean annual instructor earnings are between 157.44 and 159.36 USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.025} = 1.960$$

$$E_z = z^*(SE) = 1.960(0.488) = 0.96$$

$$\mu \in 158.40 \pm E_z = 158.40 \pm 0.96 = [157.44,\ 159.36]$$

C9: Structured Decision Procedure

$$t^* = 1.962 > 1.960 = z^*$$

$$E_t = 0.96, \quad E_z = 0.96$$

$$\text{Width}_t = 1.92, \qquad \text{Width}_z = 1.92$$

$$\text{Absolute difference} \approx 0.00 \text{ (more precisely } 0.00 \text{ to two decimals)}$$

$$\text{Relative difference (base } z\text{-interval)} \approx 0.10\%$$

For $df = 999$, convergence is very strong and both intervals are practically identical. Final decision: select the t-interval as exact; it is approximately 0.10% wider than the z-interval.

P9 Problem description An e-commerce marketplace analyzes mean monthly seller revenue (thousand USD) per store. For $n = 2500$ merchants, grouped data are:

Class (thousand USD)	m_i	f_i
200–220	210	180
220–240	230	360
240–260	250	620
260–280	270	700
280–300	290	380
300–320	310	180
320–340	330	80

1) Construct a 90% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$$\mu = \text{population mean monthly seller revenue}, \quad n = 2500, \quad \sum f_i m_i = 657000, \quad \sum f_i m_i^2 = 174410000$$

$$\bar{x} \approx \frac{657000}{2500} = 262.80$$

$$s \approx \sqrt{\frac{174410000 - 2500(262.8)^2}{2499}} = 28.784, \quad df = 2499$$

$$t^* = t_{0.05,2499} \approx 1.646, \quad SE = \frac{28.784}{\sqrt{2500}} = 0.576$$

$$E_t = t^*(SE) = 1.646(0.576) = 0.95$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 262.80 \pm 1.646 \left(\frac{28.784}{\sqrt{2500}}\right) = 262.80 \pm 0.95$$

$$\mu \in [261.85,\ 263.75]$$

Interpretation: With 90% confidence, mean monthly seller revenue is between 261.85 and 263.75 thousand USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.05} = 1.645$$

$$E_z = z^*(SE) = 1.645(0.576) = 0.95$$

$$\mu \in 262.80 \pm E_z = 262.80 \pm 0.95 = [261.85,\ 263.75]$$

C9: Structured Decision Procedure

$$t^* = 1.646 > 1.645 = z^*$$

$$E_t = 0.95, \quad E_z = 0.95$$

$$\text{Width}_t = 1.90, \qquad \text{Width}_z = 1.90$$

$$\text{Absolute difference} \approx 0.00 \text{ (more precisely } 0.00 \text{ to two decimals)}$$

$$\text{Relative difference (base } z\text{-interval)} \approx 0.05\%$$

At this scale, t and z intervals are nearly the same numerically. Final decision: use the t-interval as exact under unknown variance; it is approximately 0.05% wider than the z-interval.

P10 Problem description A streaming and creator-services network studies mean quarterly creator revenue (thousand USD) across channels. For $n = 5000$ firms, grouped data are:

Class (thousand USD)	m_i	f_i
400–440	420	320
440–480	460	680
480–520	500	1300
520–560	540	1400
560–600	580	820
600–640	620	360
640–680	660	120

1) Construct a 99% confidence interval for the population mean using the t-distribution from the grouped-data estimates. 2) Using the interval and the computed statistics, apply the structured C9 decision procedure to compare the t-based and z-based intervals and determine which interval is wider and by what percentage.

C8: Confidence Interval (t-statistic)

$\mu =$ population mean quarterly creator revenue, $\quad n = 5000, \quad \sum f_i m_i = 2631200, \quad \sum f_i m_i^2 = 1401024000$

$$\bar{x} \approx \frac{2631200}{5000} = 526.24$$

$$s \approx \sqrt{\frac{1401024000 - 5000(526.24)^2}{4999}} = 55.570, \quad df = 4999$$

$$t^* = t_{0.005,4999} \approx 2.577, \quad SE = \frac{55.570}{\sqrt{5000}} = 0.786$$

$$E_t = t^*(SE) = 2.577(0.786) = 2.03$$

$$\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) = 526.24 \pm 2.577 \left(\frac{55.570}{\sqrt{5000}}\right) = 526.24 \pm 2.03$$

$$\mu \in [524.21,\ 528.27]$$

Interpretation: With 99% confidence, mean quarterly creator revenue is between 524.21 and 528.27 thousand USD.

C5: Confidence Interval (z-statistic)

$$z^* = z_{0.005} = 2.576$$

$$E_z = z^*(SE) = 2.576(0.786) = 2.02$$

$$\mu \in 526.24 \pm E_z = 526.24 \pm 2.02 = [524.22,\ 528.26]$$

C9: Structured Decision Procedure

$$t^* = 2.577 > 2.576 = z^*$$

$$E_t = 2.03, \quad E_z = 2.02$$

$$\text{Width}_t = 4.06, \qquad \text{Width}_z = 4.04$$

$$\text{Absolute difference} = 4.06 - 4.04 = 0.02$$

$$\text{Relative difference (base } z\text{-interval)} = \frac{4.06 - 4.04}{4.04} \times 100 = 0.50\%$$

With very large degrees of freedom, the numerical gap is minimal but still positive for t. Final decision: keep the t-interval as exact under unknown variance; it is 0.50% wider than the z-interval.

Confidence Interval Summary

1 Problem 1

t-interval:

$$\mu \in [39.15,\ 42.85]$$

z-interval:

$$\mu \in [39.40,\ 42.60]$$

$$\Delta W = W_t - W_z = 0.50$$

$$\frac{\Delta W}{W_z} \times 100 = 15.63\%$$

2 Problem 2

t-interval:

$$\mu \in [1.955,\ 2.167]$$

z-interval:

$$\mu \in [1.960,\ 2.162]$$

$$\Delta W = W_t - W_z = 0.010$$

$$\frac{\Delta W}{W_z} \times 100 = 4.95\%$$

3 Problem 3

t-interval:

$$\mu \in [120.46,\ 124.54]$$

z-interval:

$$\mu \in [120.59,\ 124.41]$$

$$\Delta W = W_t - W_z = 0.26$$

$$\frac{\Delta W}{W_z} \times 100 = 6.81\%$$

4 Problem 4

t-interval:

$$\mu \in [33.04,\ 36.96]$$

z-interval:

$$\mu \in [33.08,\ 36.92]$$

$$\Delta W = W_t - W_z = 0.08$$

$$\frac{\Delta W}{W_z} \times 100 = 2.08\%$$

5 Problem 5

t-interval:

$$\mu \in [64.43,\ 66.74]$$

z-interval:

$$\mu \in [64.44,\ 66.72]$$

$$\Delta W = W_t - W_z = 0.02$$

$$\frac{\Delta W}{W_z} \times 100 = 0.88\%$$

6 Problem 6

t-interval:

$$\mu \in [113.16,\ 117.32]$$

z-interval:

$$\mu \in [113.16,\ 117.32]$$

$$\Delta W = W_t - W_z = 0.02$$

$$\frac{\Delta W}{W_z} \times 100 = 0.48\%$$

7 Problem 7

t-interval:

$$\mu \in [78.09,\ 80.15]$$

z-interval:

$$\mu \in [78.09,\ 80.15]$$

$$\Delta W = W_t - W_z = 0.01$$

$$\frac{\Delta W}{W_z} \times 100 = 0.49\%$$

8 Problem 8

t-interval:

$$\mu \in [157.44,\ 159.36]$$

z-interval:

$$\mu \in [157.44,\ 159.36]$$

$$\Delta W = W_t - W_z = 0.00$$

$$\frac{\Delta W}{W_z} \times 100 = 0.10\%$$

9 Problem 9

t-interval:

$$\mu \in [261.85,\ 263.75]$$

z-interval:

$$\mu \in [261.85,\ 263.75]$$

$$\Delta W = W_t - W_z = 0.00$$

$$\frac{\Delta W}{W_z} \times 100 = 0.05\%$$

10 Problem 10

t-interval:

$$\mu \in [524.21,\ 528.27]$$

z-interval:

$$\mu \in [524.22,\ 528.26]$$

$$\Delta W = W_t - W_z = 0.02$$

$$\frac{\Delta W}{W_z} \times 100 = 0.50\%$$

Part I

Minimal Solutions

Problem 1

$$E_t = 1.85, \qquad \mu \in [39.15,\ 42.85]$$
$$E_z = 1.60, \qquad \mu \in [39.40,\ 42.60]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 15.63\%$$

Problem 2

$$E_t = 0.106, \qquad \mu \in [1.955,\ 2.167]$$
$$E_z = 0.101, \qquad \mu \in [1.960,\ 2.162]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 4.95\%$$

Problem 3

$$E_t = 2.04, \qquad \mu \in [120.46,\ 124.54]$$
$$E_z = 1.91, \qquad \mu \in [120.59,\ 124.41]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 6.81\%$$

Problem 4

$$E_t = 1.96, \qquad \mu \in [33.04,\ 36.96]$$
$$E_z = 1.92, \qquad \mu \in [33.08,\ 36.92]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 2.08\%$$

Problem 5

$$E_t = 1.15, \qquad \mu \in [64.43,\ 66.74]$$
$$E_z = 1.14, \qquad \mu \in [64.44,\ 66.72]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.88\%$$

Problem 6

$$E_t = 2.08, \qquad \mu \in [113.16,\ 117.32]$$
$$E_z = 2.08, \qquad \mu \in [113.16,\ 117.32]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.48\%$$

Problem 7

$$E_t = 1.03, \qquad \mu \in [78.09,\, 80.15]$$
$$E_z = 1.03, \qquad \mu \in [78.09,\, 80.15]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.49\%$$

Problem 8

$$E_t = 0.96, \qquad \mu \in [157.44,\, 159.36]$$
$$E_z = 0.96, \qquad \mu \in [157.44,\, 159.36]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.10\%$$

Problem 9

$$E_t = 0.95, \qquad \mu \in [261.85,\, 263.75]$$
$$E_z = 0.95, \qquad \mu \in [261.85,\, 263.75]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.05\%$$

Problem 10

$$E_t = 2.03, \qquad \mu \in [524.21,\, 528.27]$$
$$E_z = 2.02, \qquad \mu \in [524.22,\, 528.26]$$
$$\text{Relative difference (base } z\text{-interval)} = \frac{W_t - W_z}{W_z} \times 100 = 0.50\%$$